

PolyAgora V7.8 Absolute Return Mode

Full CTO Team Specification

Based on:

- V7.6a empirical results
- V7.7 ADD-lite findings
- Dov Glickman benchmark requirements
- observed Kelly over-constraining behavior
- MRTP / Fixed Star governance architecture

Primary references:

1. Strategic Objective

The next system target is no longer:

- relative outperformance only
- better than EW
- better than momentum

The new target is:

Institutional Absolute Return Quality

Required benchmark zone:

Metric	Target
Sharpe	> 1.5
Sortino	> 1.5
Calmar	> 1.5
Correlation to SPY	< 0.35
Correlation to TLT	< 0.35
Correlation to HYG	< 0.35
Max Drawdown	ideally < 10%
Evaluation	walk-forward only
Costs	net of slippage + transaction costs

As stated in the benchmark requirements:

2. Core Diagnosis of Current System

Current architecture already demonstrates:

Capability	Status
Regime governance	strong
Drawdown reduction	strong
Runtime steering	promising
Structural fragility sensing	emerging
Alpha extraction	insufficient
Convexity preservation	partially degraded by Kelly

Interpretation:

PolyAgora is already becoming good at surviving,
but not yet sufficiently good at extracting scalable convexity.

This is normal.

Governance is harder than weighting optimization.

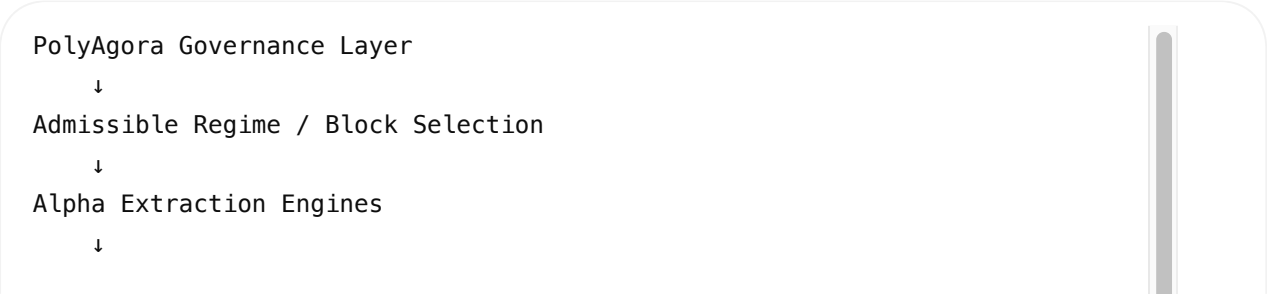
3. Fundamental Architectural Shift

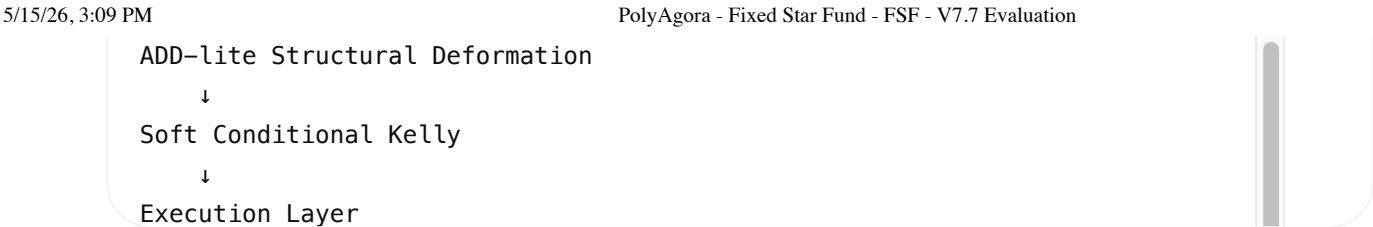
Current implicit architecture:

Signal → Kelly → Risk Reduction → Defensive Allocation

This over-flattens exposure.

New architecture:





This separation is critical.

4. PolyAgora Must Remain Governance Layer

PolyAgora should NOT become:

- another momentum engine
- another optimizer
- another factor model

PolyAgora remains:

the runtime semantic governance system.

Its role:

- determine admissible blocks
- determine recoverability
- detect fragility
- steer exposure geometry
- deform allocations under tension
- preserve maneuverability

NOT directly maximize returns alone.

5. New V7.8 Architecture

Layer 1 — Governance Layer

Existing V7.6a logic remains largely unchanged.

Responsibilities:

Function	Description
Regime classification	identify admissible state
Survival matrix	determine structural coherence
Local star/block	define admissible corridor

Function	Description
M RTP	evaluate recoverability
Transition detection	detect narrowing reachability
Runtime steering	constrain / deform / escalate

Output:

$$w_t^{base}$$

This is the base admissible allocation.

6. Layer 2 — Alpha Extraction Engines

This is the main missing layer.

Current system governance is stronger than alpha extraction.

Need independent alpha sleeves operating INSIDE governed admissible blocks.

Required Alpha Sleeves

A. Momentum Sleeve

Purpose:

- trend capture
- medium-term persistence
- convexity harvesting

Possible:

- 12–1 momentum
- adaptive momentum
- cross-sectional momentum

BUT:

must remain subordinate to governance layer.

B. Relative Value Sleeve

Purpose:

- lower correlation
- improve Sharpe
- improve market neutrality

Examples:

- spreads
 - pair structures
 - sector relative strength
 - equity vs bond asymmetry
 - carry dispersion
-

C. Volatility / Convexity Sleeve

Purpose:

- crisis alpha
- drawdown asymmetry
- Calmar improvement

Possible instruments:

- VIX structures
 - vol carry
 - convex tail overlays
 - crisis hedges
-

D. Mean Reversion Sleeve

Purpose:

- reduce downside volatility
 - improve Sortino
 - stabilize runtime
-

7. ADD-lite Redesign

Current findings strongly suggest:

ADD-lite works,
BUT Kelly interprets it too aggressively.

Observed issue:

ADD high → excessive CASH transition

This destroys CAGR.

8. ADD-lite Must Become Continuous Deformation

Critical principle from V7.8 spec:

ADD is NOT:

- crash predictor
- timing oracle
- directional predictor

ADD measures:

structural maneuverability degradation.

9. Continuous Deformation Model

Preferred Implementation

Asset-level deformation.

For each asset:

$$ADD_{i,t} \in [0, 1]$$

Then:

$$w'_{i,t} = w_{i,t}(1 - \lambda_i ADD_{i,t})$$

Preferred because:

- preserves convex winners
- avoids flattening
- localizes fragility response
- preserves asymmetry

As emphasized in the V7.8 specification:

10. ADD-lite Input Signals

Recommended components:

Signal	Interpretation
Correlation compression	crowding
Breadth deterioration	narrowing admissibility
Volatility asymmetry	latent instability
Liquidity deterioration	recoverability loss
Regime inconsistency	transition tension
Momentum saturation	convexity exhaustion

Important:

These are NOT prediction signals.

They are:

structural fragility signals.

11. Kelly Redesign

Current Kelly logic is likely over-sensitive.

Problem:

classic Kelly:

$$f_t = \mu/\sigma^2$$

becomes unstable under noisy estimates.

Result:

- excessive cash
- dead convexity
- reduced CAGR
- over-governance

12. New Kelly Logic

Kelly becomes:

conditional convexity amplifier.

NOT permanent sizing governor.

Kelly Activation Conditions

Kelly may activate only when:

Condition	Requirement
ADD low	stable admissibility
Recoverability high	MRTP healthy
Transition tension low	corridor stable
Breadth healthy	non-crowded

Suggested Kelly Gate

$$K_t = \max(0, 1 - \gamma ADD_t)$$

Then:

$$w_t'' = K_t w_t'$$

Recommended:

$$\gamma \in [0.2, 0.5]$$

NOT higher.

13. Absolute Return Objective Function

Current system should stop optimizing only Sharpe.

Use multi-objective optimization:

$$Score = 0.30Sharpe + 0.25Sortino + 0.25Calmar + 0.10CAGR - 0.10CorrToEquities -$$

As proposed in benchmark specification:

14. Walk-Forward Requirements

ALL evaluation must be:

Requirement	Mandatory
Walk-forward only	yes
No hindsight	yes
Transaction costs	yes
Slippage	yes
Turnover penalty	yes
Out-of-sample windows	yes
Rolling recalibration	yes

No retrospective fitting.

No oracle benchmark optimization.

15. Correlation Reduction Mandate

Institutional benchmark requires low beta.

Target:

Correlation	Requirement
Corr(system, SPY)	< 0.35
Corr(system, TLT)	< 0.35
Corr(system, HYG)	< 0.35

Therefore system cannot remain only directional ETF allocator.

16. Required New Structures

To lower correlation:

Structure	Purpose
spread trades	relative alpha
long/short sleeves	market neutrality
volatility sleeves	crisis convexity
carry sleeves	orthogonal returns
relative value engines	low beta
adaptive corridor allocation	MRTP steering

17. MRTP Integration

ADD should feed MRTP directly.

High ADD implies:

- shrinking reachable futures
- narrowing admissible corridors
- higher transition cost
- reduced maneuverability

Thus:

$$MRTP_t = f(Recoverability, ADD, Reachability, TransitionCost)$$

As defined in V7.8 runtime logic:

18. Runtime Philosophy

The system should evolve toward:

runtime deformation of maneuverability under changing admissibility.

NOT:

- static optimization
- static factor weighting
- static momentum
- static risk parity

But:

governed adaptive maneuverability.

19. Immediate CTO Deliverables

Phase 1

Implement:

- ADD-lite asset-level deformation
- soft Kelly gate
- transaction cost engine
- slippage model
- turnover penalties

Phase 2

Build alpha sleeves:

- momentum
- relative value
- mean reversion
- convexity / volatility

inside governed blocks.

Phase 3

Run full walk-forward evaluation:

Metric	Target
Sharpe	> 1.5
Sortino	> 1.5
Calmar	> 1.5
Correlation	low
DD	controlled

20. Final CTO Direction

The critical insight is:

PolyAgora's breakthrough is not merely prediction.

It is:

- structural governance
- admissibility steering
- recoverability preservation
- maneuverability-aware deformation
- semantic runtime control under changing regimes

The next stage is therefore:

combine this governance advantage
with institutionally competitive alpha extraction.